IE802 Notes

Projection of M

Projection of M on (x, y) space is convex hull of RI U R2

$$M_{convex} = \{x^1, \bar{x}^1, \bar{x}^2, \bar{y}^1, \bar{y}^2; x, y, \lambda^1, \lambda^2\}$$

$$\lambda^1 g_1(\frac{\bar{x}^1}{x^1}, \frac{y^1}{x^1}) \leq 0$$
$$\lambda^2 g_1(\frac{\bar{x}^2}{x^1}, \frac{y^2}{x^1}) \leq 0$$
$$x = \bar{x}^1 + \bar{x}^2$$
$$y = \bar{y}^1 + \bar{y}^2$$

$\rightarrow$ all are convex now.

perspective function

$$p(\cdot) = \begin{cases} \lambda^1 g_1(\frac{\bar{x}^1}{x^1}, \frac{y^1}{x^1}) & \lambda^1 > 0 \\ 0 & \lambda^1 = 0 \end{cases}$$

Upcoming: - Intersection cuts - Cuts for non-convex Quadratic $\Rightarrow$ Heuristic

Next Thursday 2-3 min per team short presentation Next to next -> Colorout class

Intersection Cuts

Topics to be discussed: - Intersection Cuts - Cut for non-convex quadratics - Heuristics

Intersection Cuts:

$$\min c^T y \quad Ay \leq b \quad y \in \mathbb{Z}^p$$

(i) Solve LP relaxation where we will be at corner

(2) If we drop other constraints which are not active, we end up only 2 constraints. [Diagram: A cone with a few "x" points inside, some outside, and an arrow pointing to a point inside the cone labeled "we still get some points".]

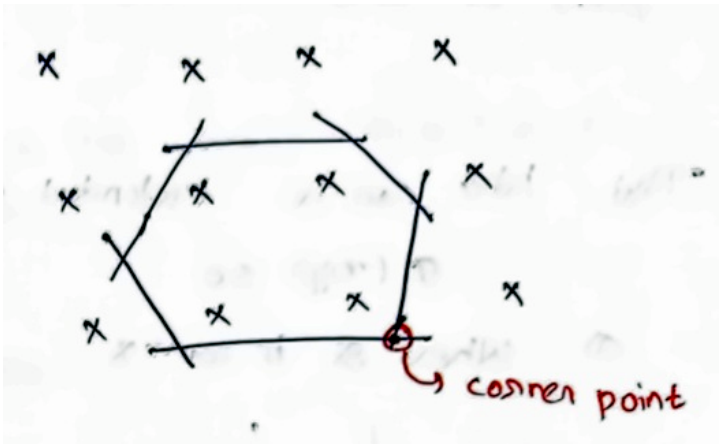


Figure 1. A polygon with vertices marked by "x"s, one red dot labeled "corner point" at the bottom right.

(3) If we look at integer points present, it is translation of cone. This is relaxation of original set.

We can chop off some points by looking at cone.

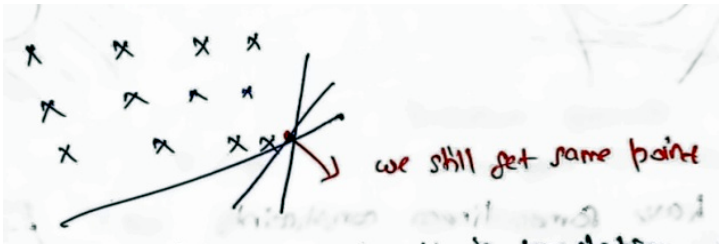


Figure 2. A cone with several "x" points inside and a few outside, an arrow from a point inside the cone to another point outside labeled "yield cut for original problem too", with a note "has to get these cuts".

We create a disjunction as all points on one of these sides leave out the points. No feasible lb hand line, all int $\rightarrow$ below or above.

- We reach from corner to corner, or point and line.

- We can choose any corner point initially, but the disjunction should be cutting the corner point (on) the corner point should be *into* the 2 disjunctions to cut-off.
- If the starting point is not a corner of feasible region but infeasible region.

This idea can be extended to non-convex too.

$$g_1(x, y) \leq 0$$

a) When g_1 is convex

b) Suppose g_1 is concave

- $m_1^2 - m_2^2 \leq -1$
- outside circle

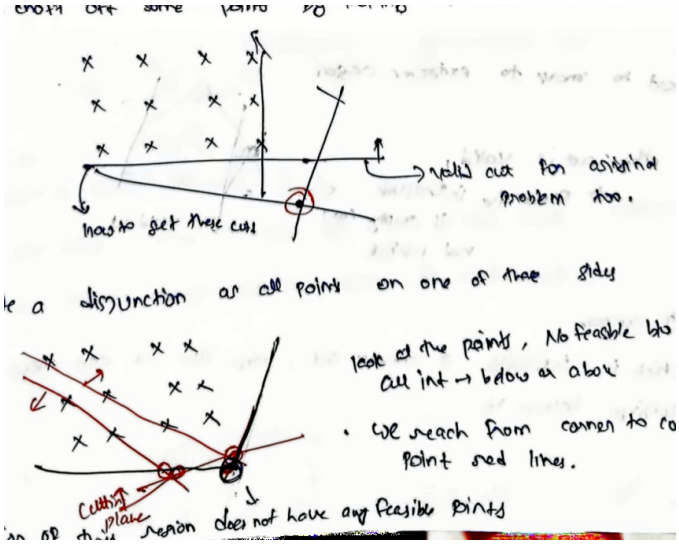


Figure 3. A region with a red cross through it, labeled "current plane", and a note "intersection of tiny region does not have any feasible point".

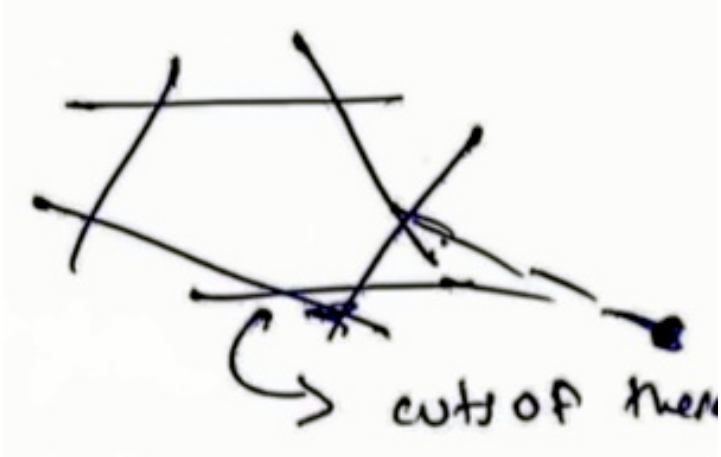


Figure 4. A polygon with a marked corner point, labeled "basic Gom point", and an arrow labeled "cut off here as much deeper".

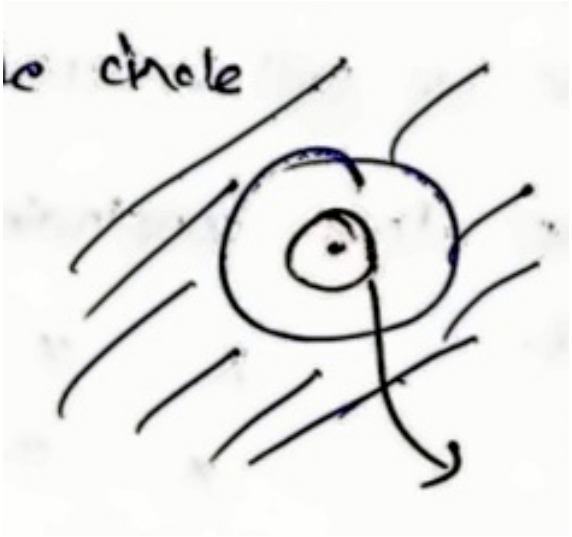


Figure 5. A square containing a smaller circle — a convex set.

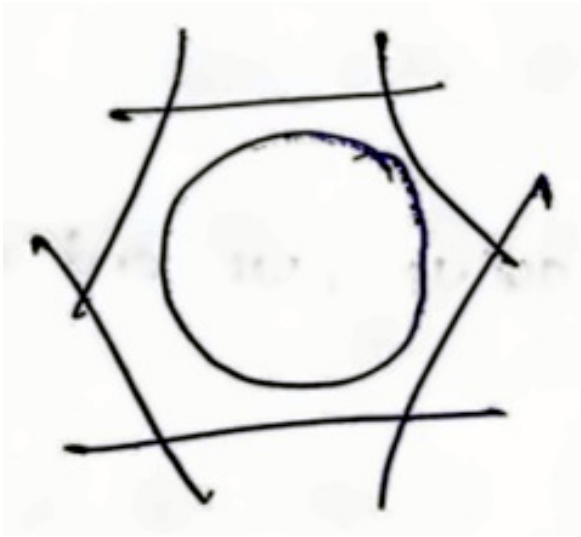


Figure 6. An irregular shape with a small circle inside, labeled “we cant chop off the palm”.

We take relaxation of this case.

- Suppose we have some linear constraints
- Since hand we have concave
- We may get corner point if it is infeasible region then we are done

If not like that we start to remove to extract region



Figure 7. A rectangle with a red dot marked on one side, labeled "not useful".

How to show the cut is valid, to prove the infeasible is empty (or) not useful

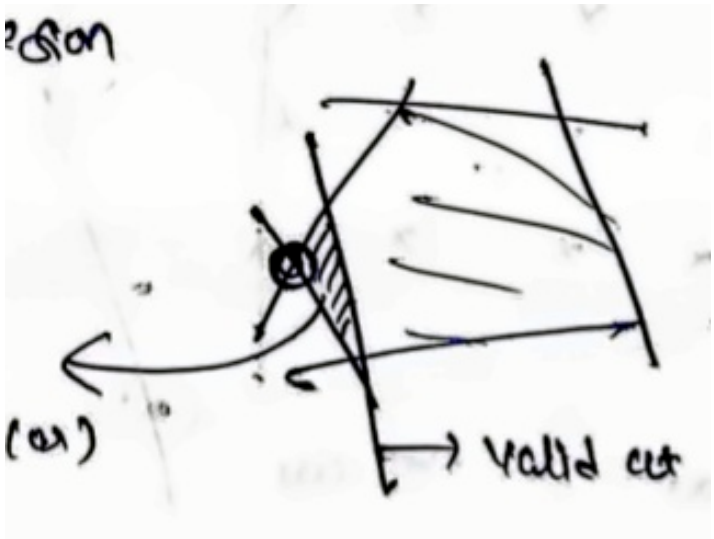


Figure 8. A shaded region with a marked point, labeled "Valid cut".

Since $g(\cdot)$ is concave

- The part which is infeasible is convex set, using this we can check there is nothing below it..

only convex for $g(x)$ is concave

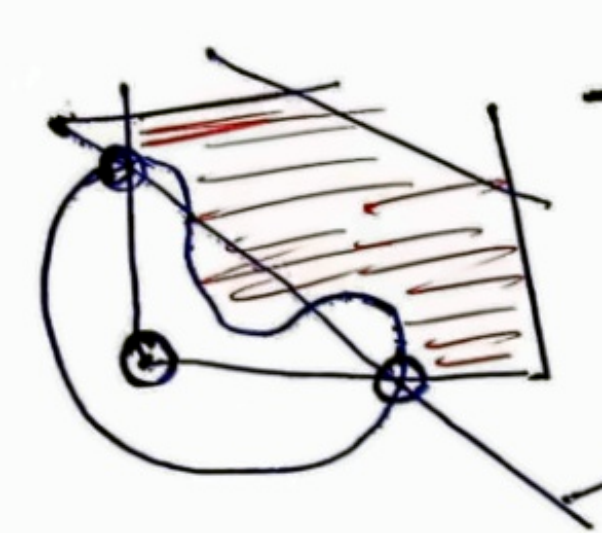


Figure 9. A shaded region with a cut through it, illustrating the idea of traveling 2 sides of a corner point and cutting all the edge of infeasible region then draw a cut.

Not a valid cut as it cuts feasible set [egon holes]

- This works for high dim.
  - 3D $\rightarrow$ 2 constraints $\rightarrow$ move 3 direction to find points $\rightarrow$ draw hyperplane that intersect & plane
  - N dim $\rightarrow$ N direction $\rightarrow$ N points provided constraint is concave if work

$$x^T Q' x \leq b_1 \quad \text{if } Q \text{ is PD it is convex}$$
$$\text{if } Q \text{ is ND} \rightarrow \text{conex}$$

If neither P or ND

DCP: $x^T(Q'^+ - Q'^-)\eta \leq b_1 \downarrow$ diff of convex fn

in ND

also as,

$$x^T Q'^+ \eta - w \leq b_1 \rightarrow \text{convex}$$

concave, $w - x^T Q'^- \eta \leq 0$

Tangent $\rightarrow$ convex Intersection point $\rightarrow$ concave

In practical using simplex of intersection point $\rightarrow$ canon plane $\Rightarrow$ many solvers do not use it unless $Q \rightarrow$ diagonal..

$$\min x^T Q x + c^T x \qquad 0 \leq x_i \leq 1 \rightarrow \text{Noninteger}$$

$\hookrightarrow$ Introduce y_i variable.

$$\min \sum Q_{ij} x_i x_j + \sum Q_{ii} y_i + \sum c_i x_i$$

$$x_{ij} = x_1 x_2 \ \forall i, j$$
$$y_i = x_i^2 \ \forall i$$
$$0 \leq x_i \leq 1$$

↪ In case of only boolean $x \in \{0, 1\}$, Padberg triangle inequalities.

$$\min \sum Q_{ij} x_i x_j + \sum c_i x_i$$

st $\quad x_{ij} = x_1 x_2 \qquad x_i \in \{0, 1\} \ \forall i$

↪ Convex hull of $\rightarrow$ is same as convex hull of $x_{ij} = x_1 x_2, 0 \leq x_i \leq 1$

Result by Burer and Letchford.

Finding Good Upper Bounds

Finding good upper bounds:

$$\min f(x, y) \quad \text{st. } g_i(x, y) \leq 0, i = 1, 2, \ldots m$$
$$x \in \mathbb{R}^n, y \in \mathbb{Z}^p$$

- Upper bound → find a feasible point as fast as possible, as low obj. value as possible → **Primal Heuristic**

Good upper bound - Prune the tree easily - Give a good solution in limited time

- Churobi may have 60-80 heuristics...

Heuristics

↪ Starting Heuristics (Don't need a feasible point to start) → Kla construction heuristics ↪ Try to construct a soln from scratch

↪ Improvement Heuristics (start with given a feasible soln ↪ find better solns)

MINLOs:

↪ Heuristic for unanchored models ↪ For model with many feasible solutions. ↪ For models with few solutions (difficult task)

> another way to classify

Heuristic ↪ General purpose (structure independent) ↪ Class specific (only Quadratic, Contin. Non-Convex Q, Conv. MIQP, conic integer models) ↪ Application Specific (portfolio opt., power flow problems, cutting stock, VRP, packing, Quad. assignment..)

Imp.

$$\min f(x, y) \quad x \in \mathbb{R}^n, \ y \in \mathbb{Z}^p$$

(1) **Algo-1**

1. Pick a random point (x, y) with $x \in \mathbb{R}^n$ and $y \in \mathbb{Z}^p$
2. Check if it is best
3. Go to 1 (until... value not improves) $\rightarrow$ stopping criteria is important works but Not so good... $\rightarrow$ we have good idea after many trials

Neighbourhood search:

(2) **Algo-2**

1. Solve continuous relaxation
2. Round y to nearest integer $\hookrightarrow$ Good if convex problem relaxation is there

Also couple with Neighbourhood search

1. Try different rounding.

we get: $y = \begin{bmatrix} 0.6 \\ 0.4 \\ 0.2 \\ 0.5 \end{bmatrix}$

$$\min f(x, y) \qquad \sum_{i=1}^{p} y_i = k, \quad y_i \in \{0, 1\},\ i = 1 \dots p$$

(1) Algo-1 $\rightarrow$ same mostly most soln on x.. if 0 or 1 not blu...

Continuous Non-convex Quadratic Models:

$$\min \frac{1}{2} x^T Q x + c^T x$$
$$\text{st } \frac{1}{2} Q_i x^T Q_i x + q_i^T x \leq b_i$$
$$l \leq x \leq u$$

0 Multistart NLP Heuristic - Give it to a local gradient based solver

also give them another point, if not good then previous take another point

0 Under cover heuristic - You pick some x values suppose p=100 $\Rightarrow$ pick 90 and fix for some values

$$x_1^2 + 5x_1x_2 + 7x_3x_4 + x_1x_3 - 2x_1x_4 \leq 1$$

ISAC Heuristic $x_1 = 2, x_3 = 1 \rightarrow$ next choose values and solve NLP (take x_1, x_2 as anything, change available)

- how to know which variable to pick and what value? $\rightarrow$ Trial and error

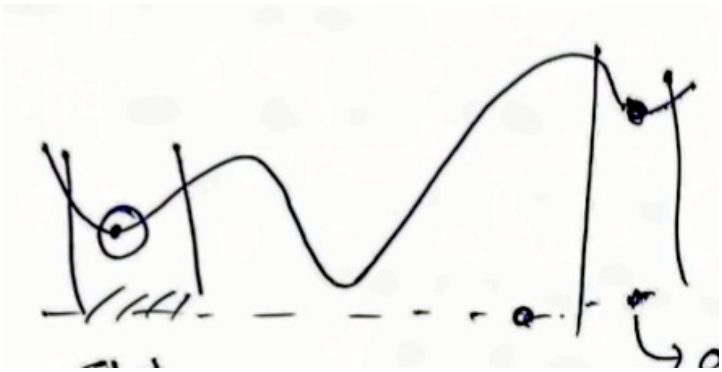


Figure 10. A wavy curve with a local minimum marked at the start, and another point further along the curve.

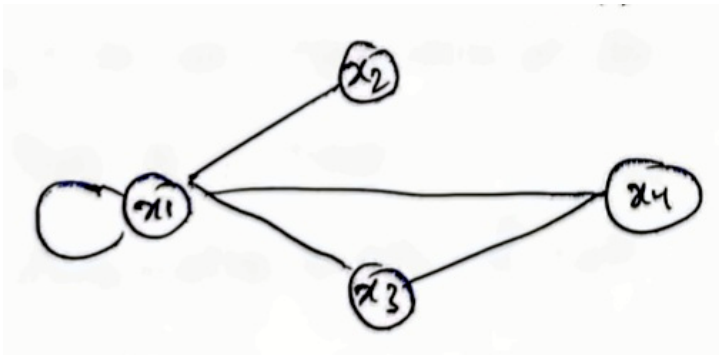


Figure 11. A graph with nodes x_1, x_2, x_3, x_4 — x_1 has edges to x_2 and x_4, x_3 has an edge to x_4.

pick smallest node so that every set is covered. - G find smallest node (edge) cover fan this graph so that every edge is covered.

(x_1, x_3) & (x_1, x_4) Solve..

Heuristics for MINLO

Initial Construction Heuristic

- Feasibility Pump (expensive)
- Rounding (sophisticated)
- Branching
- Local Branching

MZLP Case: (Pure Binary)

$$\min c^T x \quad Ax \leq b \quad x \in \{0,1\}^n$$

 Solve the LP

$$\min c^T x \quad Ax \leq b \quad 0 \leq x \leq 1 \quad \Rightarrow \hat{x}$$

↪ If fractional → round it to nearest integer

↪ Not always works

Simple rounding of $\hat{x} = \begin{bmatrix} \lceil x_1 \rceil & \text{if } x_1 < 0.5 \\ \vdots & \\ \lceil x_k \rceil & \text{on } \lceil x_k \rceil \end{bmatrix}$

$$\min \|x - \hat{x}\|_1, \quad x \in \{0,1\}^n, \ Ax \leq b$$

This will always give integers but hard...

$\square$ $\min_{i=1}^{n} |x_i - \hat{x}_i|$

$$Ax \leq b, \quad 0 \leq x \leq 1 \quad \Rightarrow \hat{x} \rightarrow \text{LP relaxation soln}$$

ex: $\hat{x} = \begin{bmatrix} 0 \\ 0.4 \\ 0.6 \\ 0.8 \\ 0.2 \end{bmatrix}$, sound($\hat{x}$) = $\begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$

$\hookrightarrow$ not working, too bad

becomes: $\min |x_1 - 0| + |x_2 - 0| + |x_3 - 1| + |x_4 - 1| + |x_5 - 1| + |x_6 - 0|$

st $Ax \leq b, 0 \leq x \leq 1$

$$\min x_1 + (1 - x_2) + x_3 + (1 - x_4) + (1 - x_5) + x_6$$

st $Ax \leq b, 0 \leq x \leq 1$

$\star$ if $\hat{x} = \begin{bmatrix} 0.1 \\ 0.9 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, again round off gives same gain $\rightarrow$ we have to stop heuristic and flip some variables.

Convex MINLO Case (Pure Binary)

$$\min c^T x + d^T y$$
$$g_i(x, y) \leq 0 \quad i = 1, \ldots, p$$
$$x \in \mathbb{R}^n, \quad y \in \{0,1\}^p \rightarrow \text{continuous}$$

$\square$ Solve NLP relaxation $\Rightarrow$ get $\hat{x}, \hat{y} \rightarrow$ round of $\Rightarrow$ you can also solve MILP relaxation

$\square$ $\min \sum_{i=1}^{p} |y^i - \text{round}(\hat{y}^i)|$ st $g_i(x, y) \leq 0, 0 \leq y \leq 1$

$\Leftrightarrow$ MIP: $\min \sum |y^i - \text{round}(y^i)|$ st ∇ inequalities at $(\hat{x}, \hat{y}), 0 \leq y \leq 1$

$$\Rightarrow \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}, \qquad y \in \{0,1\}^p$$

Non-convex MINLO Case (Pure Binary)

$\Rightarrow$ Leave it...

RINS Heuristic (Relaxed-Induced Nbd Search) → Improvement

Convex MINLO: Let $(\hat{x}, \hat{y})$ be known solution of MINLO

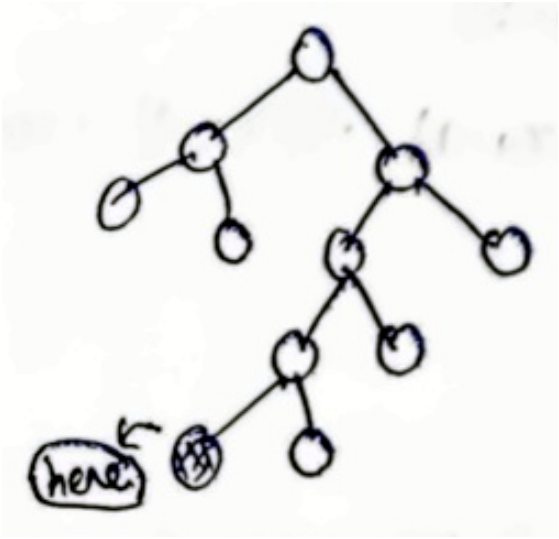


Figure 12. A tree with root node branching into two children, each of which branches further.

- Solve NLP relaxation of that node $\Rightarrow (\hat{x}, \hat{y})$, $\hat{y} \notin \mathbb{Z}^p$ → if majority of equal, randomly fix $(\hat{y}_i = \bar{y}_i)$
- comp. $\hat{y}$ and $\bar{y}$ → all those same → fix it
- Solve the smallest MINLO → don't solve fully but time limits, ex. 5 sec., and some other parameters.

Local Branching (Improvement)

Let $(\hat{x}, \hat{y})$ be a known solution, $\hat{y} \in \mathbb{Z}^{n_1}$

$$p \sum_{i=1}^{n_1} y_i - \hat{y}_1 \leq 5 \quad \text{(cardinality constraint)}$$

LOCAL JUMP: Solving LPs takes too long.

Idiot Crash:

$$\min c^T x, Ax \leq b, x \in \mathbb{Z}^p$$

$$\min \sum w_i (b_i - a_i y)^+$$

Initial Construction:

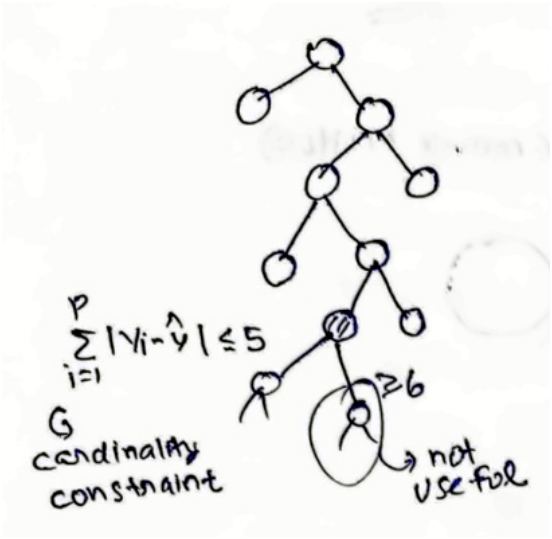


Figure 13. A branching tree — root splits into two children, each of which splits further, one branch ending at a node labeled "not useful" with $y = 6$, violating cardinality constraint.

$$2y_1 + y_2 - y_3 \leq 7 \rightarrow y_1 \text{ should be small}$$
$$-y_1 + 2y_2 + y_3 \geq 9 \rightarrow y_1 \text{ should be small}$$

$$y_2 + y_0 + y_4 = 5$$
$$y_3 - y_4 + y_6 \leq 8$$
$$0 \leq y \leq 10$$

set $y = 0$

Recall:

16-04-26

- Extended cutting plane (convex MINLO)

convex MINLO - ECP - Queveda Crossmann - Dusan and Grossmann - Non-linear branch and bound - Gurobi

- Separability
- Perspective reformulation ✓
- Automatic reformulation ✓
- Bound tightening
- Heuristics

  - Local Bb
  - RINS
  - Res Pump
  - Under Cover

- SDP relaxation
- Boolean Quadratic Polytope
- Cuts → MIR, CG, split cuts
- Secant?
- McCormick Zona
- RL7
- SOCI?
- Loth digl...